ERIC QIN

Student - Builder - Athlete

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SELECTED PROJECTS

ClimbIQ

Creator & Engineer

Built a browser-based, local-first speed-climbing video analysis platform with React, TypeScript, MediaPipe Pose, Canvas, Web Audio, and WebAssembly. Combined timing-light, audio, and athlete-motion evidence into reviewable race events and split information, with frame inspection, marker correction, local attempt saves, comparison, and export.

Evaluated 11 source recordings and 30 controlled variants; performed 126 frame seek/pixel checks and 230+ noisy-state automated trials. Tested compression, resizing, exposure, frame-rate changes, missing audio, and trimming. General timing accuracy has not been established.

github.com/ericqin0816/ClimbIQ

PolyShield

AI-agent security project

Project overview: a security layer between AI agents and MCP tools, with tool-level policies, human approval, credential isolation, a local runner, Slack alerts, and auditability. Individual contribution is not specified.

Residential Solar + Battery Economics Model

Research in development

Developing a model of when residential solar and battery storage become economically worthwhile for a Texas homeowner, using costs, electricity rates, home-energy inputs, and system performance. Results are not yet established.

Barcode Attendance Scanner

Internal tool

Built a Student Council attendance workflow using a USB barcode scanner, Chromebook, Google Sheets, and Apps Script. Matched IDs to member rows and meeting-date columns; handled duplicate scans, unknown IDs, row creation, and cell indexing.

CalcQuest

Artist / Game Design

Designed 3 original enemy concepts for an educational RPG. Collaborate with developers on character iteration, visual consistency, and expanding the enemy roster. Additional assets are in development.

QuantBot

Contributor

Contributed to a quantitative market-research project. Project features include discovery, data, signals, AI research, Discord integration, backtesting, and paper-trading research; team features are not attributed individually.

EDUCATION & ATHLETICS

Expected graduation: 2028 | GPA: 4.70 | Class rank: 32 / 885
Coursework: AP Computer Science A, AP Computer Science Principles, AP Calculus BC.
USA Climbing Youth Nationals Qualifier - Speed; personal best: 10.51 s. Varsity swimming and a competitive swimming background.

ERIC QIN

Product, work & community

PRODUCT & WORK

Worktree

Co-founder - Marketing / Product

Focus on user discovery, outreach, positioning, feedback, and product strategy for coordination between human developers and AI coding agents. Spoke with 20+ prospective users about coding-agent workflows, coordination problems, and context fragmentation; these were conversations, not product tests. Connected with a YC-backed founder for product/startup feedback.

tryworktree.com

CZT Energy USA

Intern

Supported quality-control workflows and operations, inventory, purchasing/expenses, employee records, and document organization. Used AI and automation for repetitive documents, English/Chinese translation, PDFs, Excel, forms, and administrative workflows. Shadowed strategic marketing and executive administrative operations.

Katy ISD

Water Safety Instructor - Paid role

Taught swimming for 4 weeks at approximately 7 hours per day to children across ages and skill levels, from beginners to more advanced swimmers. Focused on safety, technique, communication, and instruction.

Independent Swim Instruction

Founder / Instructor

Independently ran swim lessons primarily for Chinese children and families: 20+ swimmers, 50+ lessons, and approximately $1,000 in revenue. Managed scheduling, parent communication, payments, lesson planning, and customers.

LEADERSHIP & SERVICE

Hearts for Elders

Community-service involvement

Supported event coordination, outreach, partnerships, sponsorship/company outreach, and technical/organizational work. Organization scale during involvement included 150+ volunteers, 800+ seniors served, 10+ chapters, 10+ events, and 20+ senior homes. Personally contacted 20+ organizations and 20+ companies for collaboration, partnerships, or sponsorship-related outreach.

Student Council

E&E Officer

Supported council activities, meetings, service/cleanup, and organization. Built the barcode attendance scanner described in the project section.

SKILLS

Development: TypeScript, JavaScript, Java, React, Next.js, Vite, Google Apps Script, Git/GitHub.
Web & infrastructure: Vercel, Supabase.
AI & automation: AI-assisted development; workflow, document, and translation automation; AI-agent/MCP concepts.
Business & operations: user discovery, outreach, positioning, operations, quality control, inventory, scheduling, Excel/spreadsheets.